

Wk3.2 Ch3 Mult linear regression

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Recap: $y = f(x) + \epsilon = (\beta_0 + \beta_1 x) + \epsilon$
 $f(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$
 via $\min \text{RSS} = \sum e_i^2 = \sum (y_i - \hat{y}_i)^2$
 $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
 $\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$ *formulas*
 SI: ① 95% CI: $\hat{\beta}_i \pm z \text{SE}(\hat{\beta}_i)$
 ② HT: $H_0: \beta_i = 0$ (t-score) $t = \frac{\hat{\beta}_i - 0}{\text{SE}(\hat{\beta}_i)} \sim t_{df= n-2}$
 $H_a: \beta_i \neq 0$

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Fit: 2 ways ① $\text{KSE} = \frac{1}{n-2} \sum (y_i - \hat{y}_i)^2$ (SE(e_i))
 ② $R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}$
 $\text{TSS} = \sum (y_i - \bar{y})^2$ Total variance (of y)
 $\text{RSS} = \sum (y_i - \hat{y}_i)^2$ unexplained (of $\hat{y}$)
 $\text{TSS} = \text{RSS} + \text{SSR}$
 $\text{SSR} = \sum (\hat{y}_i - \bar{y})^2$ Explained (of $\hat{y}$)
 $R^2 = \frac{\text{SSR}}{\text{TSS}}$ (% variance explained)
 $R^2 = \text{Corr}(X, Y)^2 = r^2$ only for SLR

Extend to Multiple LR: $X = (x_1, x_2, \dots, x_p)$
 $y = f(x) + \epsilon = (\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p) + \epsilon$
 $\vec{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$
 $\vec{x} = (1, x_1, \dots, x_p)$
 Use data to fit $f(x) \approx \hat{f}(x) = \vec{\beta}^T \vec{x}$
 $\vec{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ *no nice $\hat{\beta}_i$ formulas*

(Matrix form of $\vec{\beta}$) Ordinary Least Squares
 $X \vec{\beta} = \vec{y}$ no soln $\rightarrow X^T X \vec{\beta} = X^T \vec{y}$ normal eqn
 $\vec{\beta} = (X^T X)^{-1} X^T \vec{y}$

Model fit ($f(x) \approx \hat{f}(x)$ quality)
 2 ways: ① $\text{KSE} = \sqrt{\frac{1}{n-(p+1)} \sum (y_i - \hat{y}_i)^2} = \sqrt{\frac{1}{n-(p+1)} \text{RSS}}$ (same as SLR)
 ② $R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = \frac{\text{SSR}}{\text{TSS}}$ (% var explained)
 $= \text{Cor}(Y, \hat{Y})^2$ *product only one param.*

Inference 2 views: ① Individual β_i 's, same as SLR
 95% CI $\hat{\beta}_i \pm z \text{SE}(\hat{\beta}_i)$
 HT: $H_0: \beta_i = 0$ $t \rightarrow$ score
 $H_a: \beta_i \neq 0$ *Fisher*

② Entire model fit (F-test)
 $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$ (no X-Y rel)
 $H_a: \text{At least one } \beta_i \neq 0$
 $F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)} = \frac{\text{SSR}/p}{\text{RSS}/(n-p-1)}$ *Model variance expl*
 $F \approx 1$, model low expl variance
 $F \gg 1$, model high expl variance
 $F \sim F_{p, (n-p-1)}$ (F-dist) & gives a p-val.

MLR nuances ① How to tell if all β_i 's matter?
 Adding more features ($x_{p+1}, x_{p+2}, \dots$) always help (increase)
 KSE & R^2 , so a small fix adjusted R^2
 R^2 penalty for more features
 Goal: Find "best" feature set. Try many models & see which features have no improvement.
 $x_1, x_2, \dots, x_p$ (2^p total models) *too many!*
 3 approaches for variable selection:
 ① Forward: Add one variable at a time which has best improvement, stop when no/little change
 ② Backwards: Start with all x_i 's, delete one with least value of interest
 ③ Hybrid of ① & ②

② Multicollinearity (some x_i 's may be correlated)
 Ex: $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = \beta_0 + \beta_1 x_1 + \beta_2 c x_1$ *OLS*
 $\vec{\beta} = (X^T X)^{-1} X^T \vec{y}$ *nearly singular*
 β_1 & β_2 mixes & values meaningless
 Expect $\text{Cor}(x_1, x_2) \approx 1$, if $x_2 = c x_1$
 Unstable $\hat{\beta}_1$ & $\hat{\beta}_2$ values & t-tests provide
 Check for multicol via $VIF = \frac{1}{1 - R_j^2}$
 Loose rule: $VIF \approx 1$ ok, $VIF \approx 5$ $\odot$, $VIF \approx 10$ $\odot\odot$
 $x_j \approx \hat{f}_j(x_1, x_2, \dots, x_p)$ not x_j

2 Extensions for MLR:
 ① Polynomial MLR: If $f(x)$ not linear, try quad, cubic, n^{th} deg poly Related
 Assume $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n$ *many other data transformations*
 $\ln(x) = p \ln(x)$
 Basis-variance tradeoff (Ch2)
 ② Interaction terms: $Y = (\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2) + \epsilon$
 $\frac{\partial f}{\partial x_1} = \beta_1 + \beta_3 x_2$ *soft & harder has β_3 effect*
 soft inc in price